

SULABH KATILA

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(646) 548-8328 New York, NY

EDUCATION

St. Joseph's University

B.S. in Mathematics & Computer Science, Minor in Data Science

May 2026

Valedictorian | GPA: 3.95/4.00

TECHNICAL SKILLS

Languages: Java, Python, C#, Swift, Bash, TypeScript, SQL

Frameworks: .NET, Pandas, NumPy, FastAPI, Scikit-learn, PyTorch, Selenium, React, Express, Next.js

Tools: Unix/Linux, Git, Docker, Google Cloud Platform (GCP), NodeJS, MySQL, Azure DevOps/Pipelines

EXPERIENCE

TransPerfect

New York, NY

Software Engineering Intern (QA Automation)

February 2025 – May 2026

- Designed and developed custom code-analysis tool using Python to automatically update 400+ C# test methods, saving 80+ hours
- Built NLP evaluation pipelines to benchmark 5 large language models across 9 task categories, surfacing performance gaps that informed the production model selection decision
- Contributed to development of Model Context Protocol (MCP) integrations that enabled coding agents to securely retrieve documentation, query internal resources, and execute structured tool calls as part of an agentic development workflow
- Developed and maintained automation framework using C# & .NET to validate complex web application workflows
- Refactored legacy automated test suites plagued by flaky failures, cutting the false-positive rate by 65%, reducing CI/CD pipeline execution time by ~30 minutes per run, and lowering overall infrastructure resource usage

St. Joseph's University

New York, NY

Tutor

September 2023 – May 2026

- Tutored college Mathematics and Computer Science courses, including Advanced Programming in Java, API design, and Database

Summer Undergraduate Research Fellow (SURF)

June 2024 – August 2024

- Conducted intensive exploratory data analysis using Python, Pandas, NumPy, and Matplotlib, interpreting trends and visualizations to identify borough-level market dynamics and translate statistical patterns into actionable research insights
- Built and compared machine learning models in Scikit-learn and TensorFlow, critically evaluating performance differences to determine which architectures best captured the nonlinear forces driving NYC real estate fluctuations
- Delivered an end-to-end independent research pipeline and presented findings to an academic audience at a research symposium

HealthStream

Remote, USA

QA Engineering Intern

September 2024 – December 2024

- Established automated regression testing for a web application with 650,000+ monthly active users by building a Python/Selenium framework from scratch, providing a scalable foundation for long-term test automation

Boys Hope Girls Hope of New York

New York, USA

Mathematics Tutor

February 2024 – May 2024

- Tutored Mathematics to high school students from underserved backgrounds, providing both one-on-one and group instruction

PROJECTS

BatMapper iOS:

- Built an app that reconstructs 2D indoor floor plans by fusing sound signals collected from microphone with IMU (accelerometer + gyroscope + magnetometer) sensor data as the user walks around
- Implemented an acoustic signal processing pipeline (bandpass filtering, cross-correlation, peak detection) to extract real-time wall distances from audio, feeding a Kalman filter and dead-reckoning based algorithm to localize the user and map the room

Smartphone Placement Detection:

- Built a real-time system that detects phone's location (pocket, hand, or bag) using its motion sensors, achieving 93.4% accuracy
- Built a Swift iOS app streaming live accelerometer and gyroscope data to a Python FastAPI backend running on-the-fly feature extraction to feed to a ported MATLAB-trained AdaBoostM2 model, with predictions broadcast via WebSocket

Programming Language:

- Engineered a performant Lox programming language interpreter in Rust, averaging 65% of Python 3.12.4's execution speed

HONORS & AWARDS

- Winner, HackNYU 2025
- Hon. Mention, CalHacks 12.0
- 8 x Dean's List, SJNY
- Kappa Gamma Pi
- Delta Epsilon Sigma
- Esse Non Videri Scholarship, SJNY